OPPONG KWABENA ADUTWUM

MACHINE LEARNING ENGINEER

Applied ML · Robotics · Embedded Intelligence · Full-Stack Development

CONTACT

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TECHNICAL SKILLS

Programming Languages
- Python
- JavaScript
- C++

Data & Analytics
- Pandas / NumPy
- SQL / PostgreSQL
- Jupyter

Web & Backend
- FastAPI / Flask
- React Native / Django

Dev Tools
- Git / GitHub
- VS Code
- Linux (Kali)

Core Knowledge
- Machine Learning (fundamentals)
- Network Security
- Embedded Systems / IoT
- PID Control / Robotics

LANGUAGES

- English — Native
- French — Intermediate

CERTIFICATIONS

- Hashgraph Developer Course — Hedera (Jan 2026)
- Hedera Certified Foundation (Apr 2026)
- IPPF Competitor — Brewer Foundation & NYU (2024-25)
- WASSCE General Science — Prempeh College (Sep 2025)
- *[Deep Learning Spec.] — DeepLearning.AI*
- *[ML Specialization] — Stanford / Coursera*

GITHUB HIGHLIGHTS

- *[repo] — CNN crop disease classifier ★[X]*
- *[repo] — LLM-powered study assistant ★[X]*
- *[repo] — Sentiment analysis pipeline ★[X]*
- *[X] contributions | [X] PRs (12 mo.)*

OPEN SOURCE

- *[Hugging Face / PyTorch] — PR #[X]*
- *[scikit-learn / LangChain] — PR #[X]*

RESEARCH & BLOG

- *"[Title]" — [arXiv / Conference]*
- *"[Title]" — [Medium / Hashnode]*

INTERESTS

- Mathematics
- Movies
- Video Games
- Machine Learning

◆ PROFESSIONAL SUMMARY

Computer Science undergraduate at KNUST with hands-on experience in embedded intelligence, autonomous systems, and applied machine learning. National Robotics Champion. Regional Cybersecurity Champion.

◆ EXPERIENCE

Machine Learning Trainee — Zap Tech Technologies
Remote | May 2025
- Completed an intensive 2-week ML engineering program covering the full model development lifecycle — from raw data collection through model evaluation and iteration.
- Preprocessed and cleaned real-world datasets using Pandas and NumPy, handling missing values, feature scaling, and data augmentation.
- Trained and evaluated classification models using Scikit-learn, interpreting accuracy, precision, recall, and F1-score across iterations.

***[AI/ML Engineering Intern]** — [Company Name — e.g. Anthropic, Google DeepMind]*
[City, Country] | [Mon YYYY – Mon YYYY]
- *Trained and fine-tuned [model type] on [dataset], improving [metric] by [X]% over baseline.*
- *Built end-to-end ML pipeline: data ingestion → preprocessing → training → evaluation → deployment on [AWS/GCP].*

***[ML Research Assistant]** — [KNUST AI Lab / Dept.]*
Kumasi, Ghana | [Mon YYYY – Present]
- *Researched [topic, e.g. NLP for low-resource African languages] under Prof. [Name].*
- *Implemented [model/algorithm] in PyTorch, achieving [X]% improvement on [benchmark].*

◆ EDUCATION

Kwame Nkrumah University of Science & Technology (KNUST)
B.Sc. Computer Science | Kumasi, Ghana | Expected 2029

Prempeh College
WASSCE — General Science | Kumasi, Ghana | Graduated 2025

◆ PROJECTS

[Computer Vision Project — e.g. Crop Disease Detector]
[PyTorch · CNN/ResNet · FastAPI · AWS] | [github.com/...] [live-demo]
- *Trained [CNN/ResNet/YOLO] on [X]K images achieving [X]% accuracy on [dataset]. Solved [real-world problem].*
- *Built inference pipeline: image upload → preprocessing → prediction → confidence score in [X]ms.*
- *Deployed via FastAPI on [AWS/GCP], serving [X] predictions/day. THIS is what Anthropic/Google want to see.*

[NLP / LLM Project — e.g. AI Study Assistant]
[Claude/OpenAI API · LangChain · RAG · Next.js] | [github.com/...] [app-url]
- *Built RAG pipeline: document ingestion → embedding → vector search → LLM generation with [X]% relevance.*
- *Implemented prompt chaining and guardrails for safe, grounded responses. [X] queries/day.*

[Predictive ML Project — e.g. Energy Demand Forecasting]
[Scikit-learn · XGBoost · Pandas · Streamlit] | [github.com/...] [live-demo]
- *Built [regression/classification] model with [X]% accuracy. Feature engineering from [X] raw features.*
- *Deployed interactive dashboard on Streamlit for real-time predictions.*

Arethos — Mental Wellness App (3-Person Team)
React Native · Expo SDK 54 · SQLite · Groq & Anthropic AI | github.com/Oppong-py/arethos
- Co-built a cross-platform mental wellness app for KNUST students with AI-powered journaling, mood tracking, and reflections.
- Engineered dual-AI fallback (Groq + Anthropic), custom haptic feedback calibrated through 9 iterations, and offline-first SQLite storage.

Autonomous Line-Following Robot
LEGO MINDSTORM EV3 · PID Control Algorithm
- Designed PID control algorithm (a core ML precursor) for autonomous navigation; National Champion at Robofest Ghana 2025.
- Optimized sensor-motor feedback loops — directly applicable to reinforcement learning and robotics control.

Class Attendance Monitoring System

React Native · Django · PostgreSQL · Bluetooth

- Architected biometric attendance platform serving 50 students in prototype testing, reducing manual roll-call by ~80%.
- Built Django REST API handling concurrent Bluetooth fingerprint scans — experience in data ingestion at scale.

Solar-Powered IoT Watering System

Arduino · IR Sensor · C++ · Solar

- Designed sensor-driven embedded system with proximity-triggered actuation — edge computing for IoT.
- Engineered full circuit with solar charge controller, relay logic, and sensor fusion.

◆ HACKATHONS & COMPETITIONS

National Cybersecurity Challenge — Regional Champion (2024)

- Competed in a knowledge-based quiz testing cybersecurity fundamentals, threat analysis, and security best practices.

Robofest Ghana — National Champion — Robotics (2025)

- Built autonomous robots under timed conditions to complete designated tasks; designed and assembled within strict time limits.

ACES CodeFest 2026 *— UI/UX Challenge — Upcoming* (July 2026)

- *Competing in the ACES KNUST CodeFest UI/UX challenge — redesigning the ACES website into a mobile-first experience across 9 pages in a 2-week sprint.*

◆ LEADERSHIP & VOLUNTEERING

Club Leader — Prempeh College Cybersecurity Club (2023–2025)

Participant — International Public Policy Forum (IPPF) (2024)

Volunteer — Solutions for Life Initiative Ghana (SFLIG)

◆ MEMBERSHIPS & COMMUNITIES

Member — Prempeh College Robotics Club (2023–2025)

Member — AmaliTech Coding Club (2025–Present)

Member — Microsoft User Group Ghana — KNUST Chapter (2025–Present)

Member — Claude Builder Club — KNUST (2025–Present)